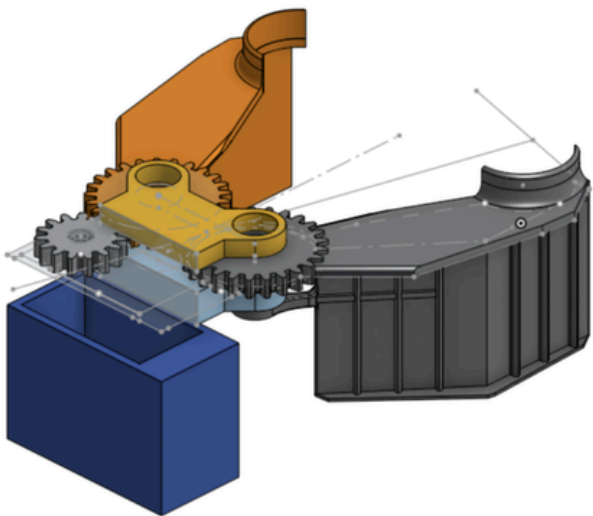
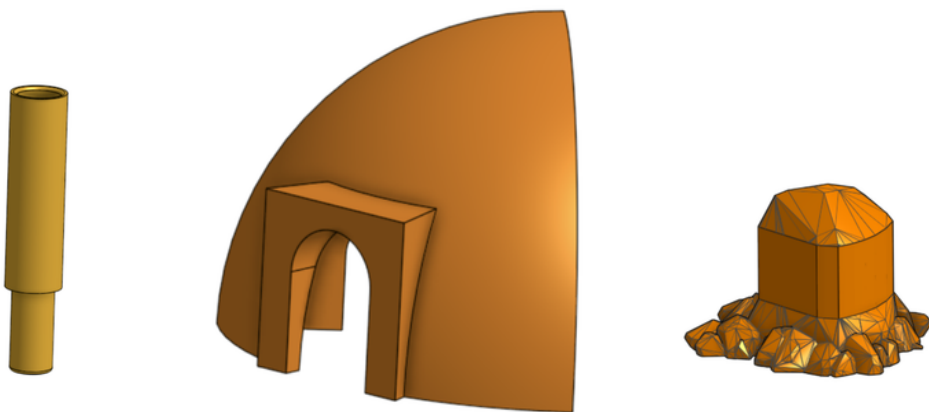
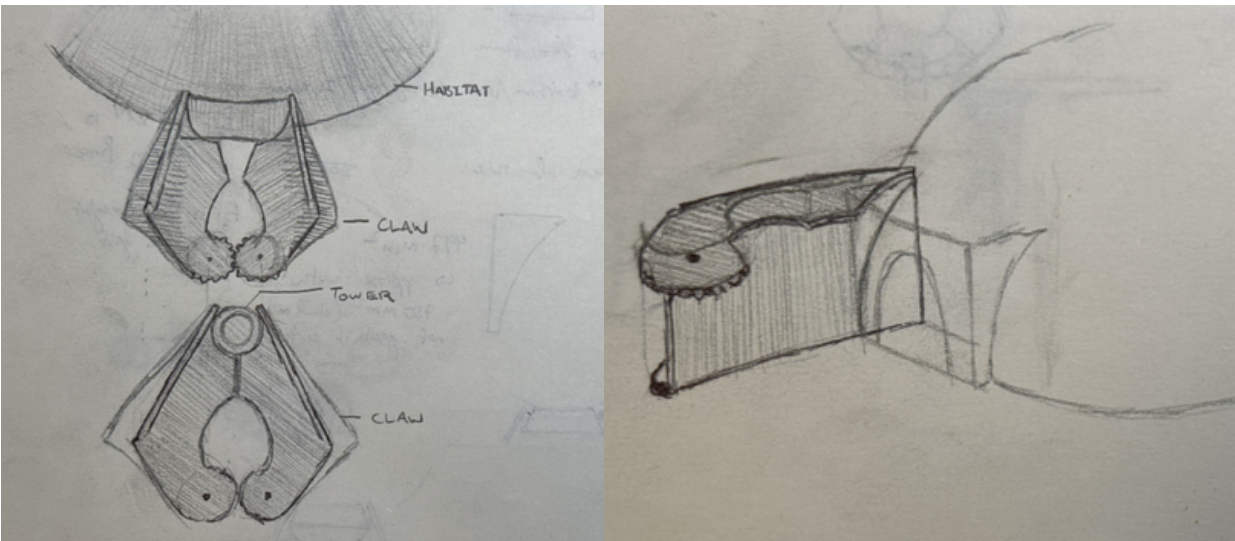


AUTONOMOUS CLAW

3D Printing | Rapid prototyping | Real-world constraint analysis | Testing and validation

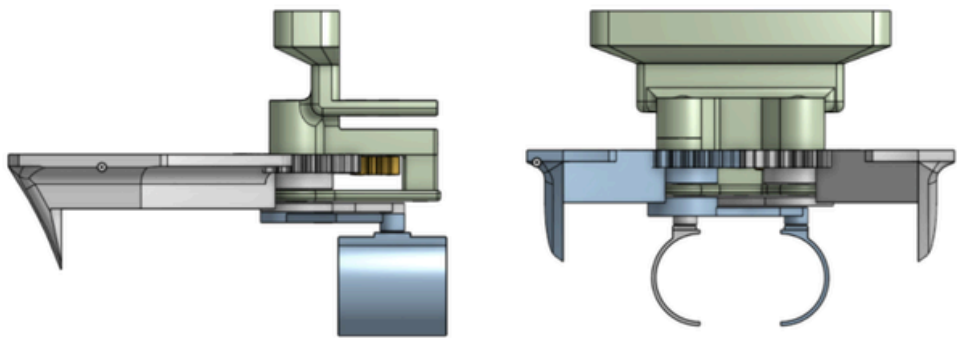
This claw design was one of my tasks out of a larger autonomous robot project. The objective of the robot was to complete a set of tasks meant to replicate/simulate an autonomous Mars robot habitat setup, in a competition setting. This robot used things such as machine imaging, tape following, and sonar.

I analyzed the geometries and weights of the various objects (shown below), and iterated on designs which focused on multifunctionality.

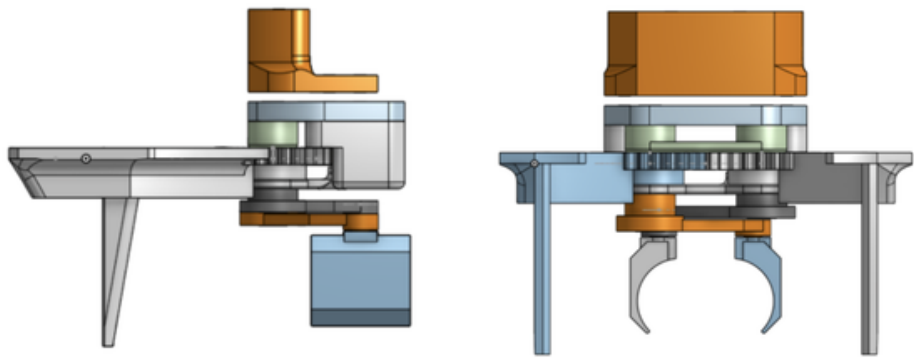
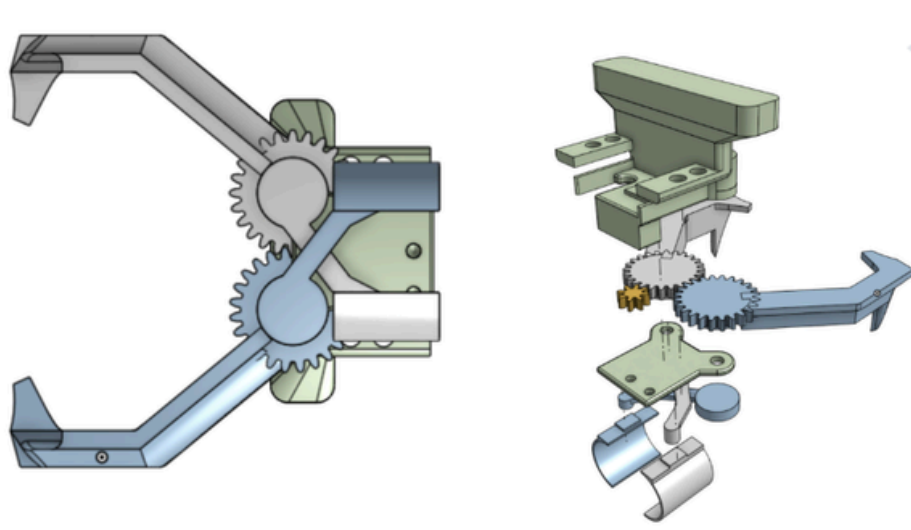


This is a model of my first design. It has all the features and a selected gear ratio, but is big and clunky. Hard to manufacture.

This is my second design, focusing on manufacturability. It still achieves the same function as the first claw but is now 3D printable with minimal supports. The mechanism for the tower pieces is also altered to accommodate for a change in our overall team strategy.



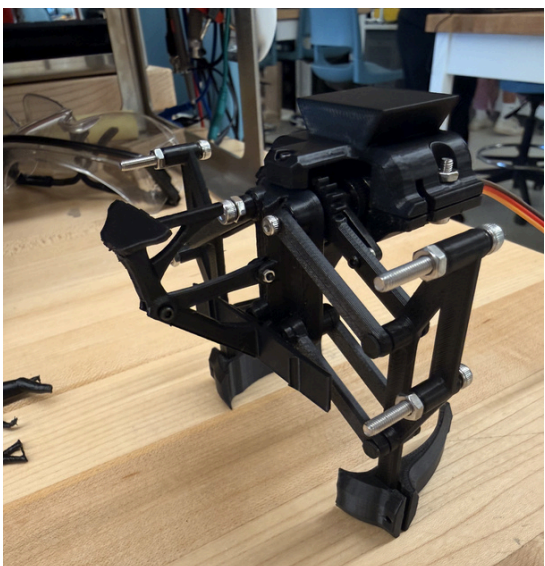
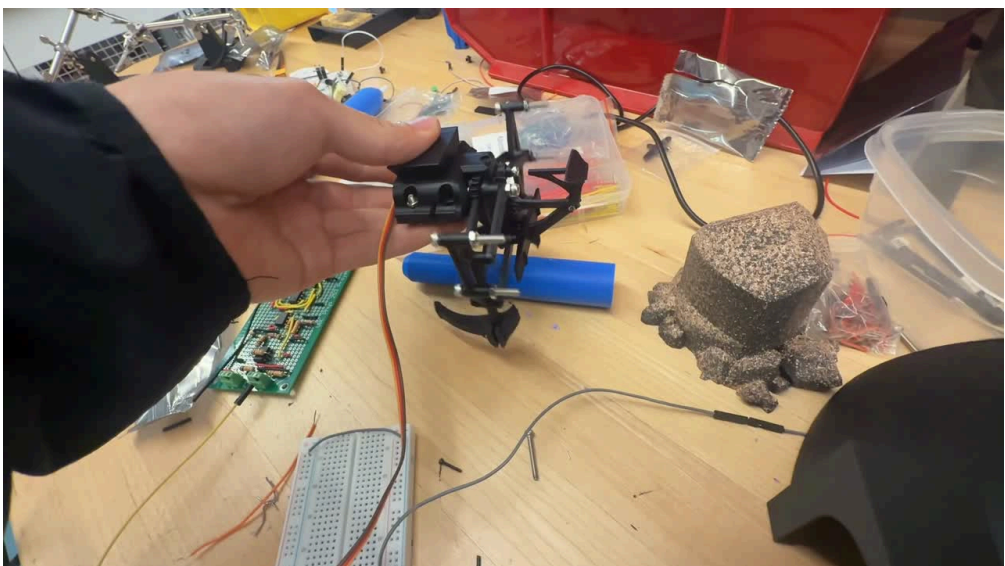
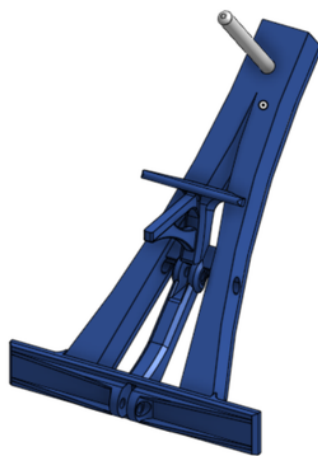
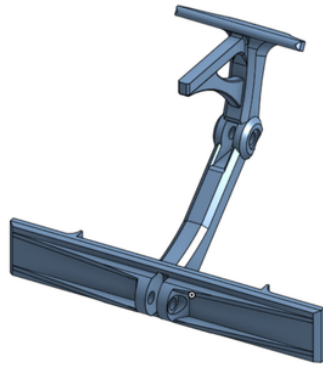
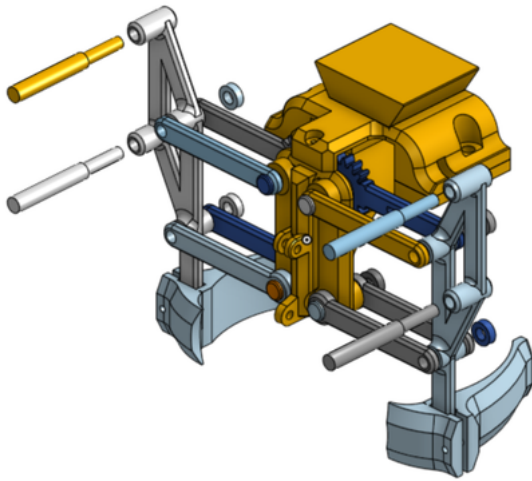
After testing, we realized that the claw wasn't as good at stopping the outward rotation of the habitat as we thought. So I changed the claw piece and tested it again.



it works, but requires a very precise placement. This would make the control side of our robot unnecessarily difficult. Our project time line was on track, so I explored another avenue for the design with all of the intuition I built off of physical testing.

4 bar linkage. Offload weight of the largest piece to a 'horn'. This also meant we could reduce the servo size to draw less power in operation.

Two horn designs; second one fixes the issues of only being able to approach the doorframe head-on. With the second one we can approach from an angle and tilt the claw up to auto-align the habitat piece to the crossbar.



More mechanically complex, but more reliable and is more power efficient - all while achieving the same tasks.

[BACK](#)